comparison between multiple models because all models with a long context length c' need to improve over the same baseline. Sometimes we only care about those positions where the baseline performs poorly (which means short-term dependency with context length c is not sufficient), so given a ratio parameter r, we define the set $\mathcal{T}$ is the above equation as

$$\mathcal{T} = \text{top-}r \text{ positions } t \text{ with largest } b(c, t)$$

The *relative gain* is subsequently defined as the relative perplexity reduction:

$$g_i(c, c') = \frac{\exp f_i(c, c) - \exp f_i(c, c')}{\exp f_i(c, c)}$$

Given a step size Δ, we then use an algorithm to find the RECL by thresholding the relative gain:

1. Set initial short context length c, and long context length $c' = c + \Delta$
2. Compute $g_i(c, c')$. If $g_i(c, c') < 0.01$, return RECL $= c$. If $g_i(c, c') \geq 0.01$, set $c = c', c' = c + \Delta$ and go to step 1.

In Figure 3, we visualize the unnormalized relative perplexity gains $(\exp f_i(c, c) - \exp f_i(c, c'))$ with various pairs of (c, c') when $r = 0.1$. It is clear that Transformer-XL has a longer RECL compared to RNNs and other baselines because the relative gains are substantially larger.

For reference, we plot the perplexities with varying context lengths in Figure 4. The y-axis denotes the "normal" perplexity (not calibrated by baselines).

D Attention Visualization

In this section, we provide some visualization of the attention learned by the SoTA model on the WikiText-103 validation set. Recall that, this model has 16 10-head transformer layers and relies on a memory of length 640.

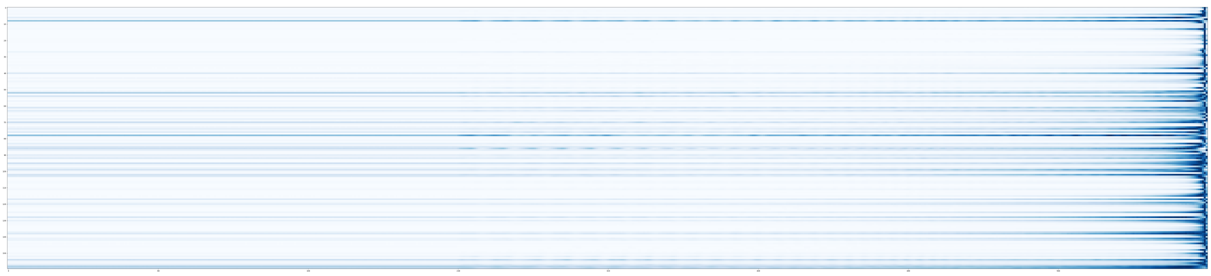

Figure 5: Average attention over the previous 640 tokens, where each row corresponds to a attention head and each column corresponds to a relative location. There are totally 160 attention heads, and every 10 heads come from a single layer. Darker colors indicate higher values.

The first visualization aims at revealing the overall trend of where the model is attending. Specifically, for each attention head of each layer, we average the attention distributions of all tokens in the validation set. This is shown in Fig. 5. As we can see, the overall trend is to focus more on the nearby tokens than the faraway ones. However, it is also very clear that some attention heads have a wider attention distribution over the entire memory span, notably the head 8 from layer 1, head 78 from layer 8, and the head 158 from layer 16.

Since we are focused on learning long-range dependency, we are especially interested in these heads with a wider attention span. Thus, in the second set of visualization, we pick the three notable heads mentioned above, and visualize their attention behavior for a randomly chosen position, as shown in Fig. 6. Here, we see three different patterns of wider attention:

- For the head 8 in the 1st layer, we see an almost uniform attention over the entire memory span. This is quite intuitive, as lower-level layers needs to screen the entire memory span to decide where to focus for higher-level layers